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Electronic circuitry

Abstract

According to one embodiment, electronic circuitry, includes a first radiating element; a second radiating element; a first circuit element including a first end and a second end, the first end being connected to one end of one of the first radiating element or the second radiating element; and a second circuit element connected between the second end of the first circuit element and one end of another of the first radiating element or the second radiating element.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application is based upon and claims the benefit of priority from the prior Japanese Patent Application No. 2022-094722, filed on Jun. 10, 2022, the entire contents of which are incorporated herein by reference.

FIELD

(2) Embodiments described herein relate to electronic circuitry.

BACKGROUND

- (3) Wireless devices including a plurality of antennas as radiating elements suffer from a problem that realized gains of the antennas are reduced due to large electromagnetic coupling among the antennas. As a decoupling circuit reducing coupling between antennas, a configuration to reduce electromagnetic coupling between two antennas by using three circuit elements is known.
- (4) In this configuration, it is necessary to adjust the three circuit elements. Therefore, complicated work is necessary to adjust the circuit elements. For example, in a case where chip components are used as the circuit elements, adjustment work of replacing three chip components while measuring the electromagnetic coupling between the antennas is necessary. In addition, a manufacturing cost is increased.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a diagram illustrating a configuration example of a decoupling circuit as electronic circuitry according to a first embodiment;
- (2) FIGS. 2A to 2D each is a diagram illustrating configuration examples of a reactance element and a susceptance element;
- (3) FIG. 3 is a diagram to explain an example of determining circuit constants of the reactance element and the susceptance element;
- (4) FIG. 4 is a diagram illustrating another configuration example of the decoupling circuit according to the first embodiment;
- (5) FIG. 5 is a diagram illustrating frequency characteristics of coupling between antennas of the decoupling circuit according to the first embodiment;
- (6) FIG. 6 is a diagram illustrating the frequency characteristics of coupling between the antennas when a reactance of the reactance element of the decoupling circuit according to the first embodiment is changed;
- (7) FIG. 7 is a diagram illustrating the frequency characteristics of coupling between the antennas when a susceptance of the susceptance element of the decoupling circuit according to the first embodiment is changed;
- (8) FIGS. 8A and 8B each is a diagram to explain an effect obtained by the decoupling circuit according to the first embodiment;
- (9) FIG. 9 is a diagram illustrating a configuration example of a decoupling circuit according to a second embodiment;
- (10) FIGS. 10A and 10B each is a diagram illustrating a configuration example of a first matching circuit;
- (11) FIG. 11 is a diagram illustrating another configuration example of the decoupling circuit according to the second embodiment;
- (12) FIG. 12 is a diagram illustrating still another configuration example of the decoupling circuit according to the second embodiment;
- (13) FIG. 13 is a diagram illustrating a configuration example of a wireless device according to a third embodiment; and
- (14) FIG. 14 is a diagram illustrating another configuration of the wireless device according to the third embodiment.

DETAILED DESCRIPTION

- (15) According to one embodiment, electronic circuitry, includes a first radiating element; a second radiating element; a first circuit element including a first end and a second end, the first end being connected to one end of one of the first radiating element or the second radiating element; and a second circuit element connected between the second end of the first circuit element and one end of another of the first radiating element or the second radiating element.

(16) Some embodiments are described below with reference to drawings.

First Embodiment

(17) FIG. 1 is a diagram illustrating a configuration of a decoupling circuit **100** as electronic circuitry according to a first embodiment. The decoupling circuit **100** relates to a technique that reduces electromagnetic coupling (hereinafter, referred to as coupling) between two radiating elements (antennas) **101a** and **101b**.

(18) The decoupling circuit **100** includes a first radiating element **101a**, a second radiating element **101b**, a circuit element **102** (first circuit element), and a circuit element **103** (second circuit element). The circuit element **102** is an element to be mainly adjusted in reactance as a circuit constant (circuit parameter), and is also referred to as a reactance element **102**. The circuit element **103** is an element to be mainly adjusted in susceptance as a circuit constant (circuit parameter), and is also referred to as a susceptance element **103** hereinafter.

(19) Each of the first radiating element **101a** and the second radiating element **101b** is an antenna that transmits, as electromagnetic waves, a high-frequency signal that is a radio signal to a space, or outputs, as a high-frequency signal, electromagnetic waves received from the space.

(20) Examples of each of the first radiating element **101a** and the second radiating element **101b** include a dipole antenna, a monopole antenna, an inverted-L antenna, an inverted-F antenna, a patch antenna, a slot antenna, a notch antenna, and a dielectric resonator antenna. The type of each antenna is not limited as long as each antenna can transmit/receive the high-frequency signal.

(21) The first radiating element **101a** and the second radiating element **101b** may be antennas having the same configuration, or may be antennas having different configurations.

(22) Further, each of the first radiating element **101a** and the second radiating element **101b** may be an antenna operating at a single frequency, or may be an antenna operating in a plurality of frequency bands.

(23) The first radiating element **101a** and the second radiating element **101b** may not be matched in an operation frequency band. Each of the first radiating element **101a** and the second radiating element **101b** may be an antenna operating at a single frequency, or may be an antenna operating in a plurality of frequency bands.

(24) The first radiating element **101a** and the second radiating element **101b** may be antennas that transmit/receive the high-frequency signal in the same polarization, or may be antennas that transmit/receive the high-frequency signals in different polarizations.

(25) The first radiating element **101a** and the second radiating element **101b** may be in any position and direction. For example, the positions and the directions are determined so as to reduce a correlation coefficient of radiation patterns of the first radiating element **101a** and the second radiating element **101b**, which makes it possible to improve communication quality and the like.

(26) An end (first end) of the reactance element **102** is connected to one end of the radiating element **101a**. One end of the susceptance element **103** is connected to the other end (second end) of the reactance element **102**, and the other end of the susceptance element **103** is connected to one end of the radiating element **101b**. The susceptance element **103** is connected between the other end (second end) of the reactance element **102** and the one end of the radiating element **101b**.

(27) The reactance element **102** is positioned between a connection point **104a** between the susceptance element **103** and the reactance element **102**, and the radiating element **101a**. The other end of the reactance element **102** is connected to a port 1 receiving/outputting a signal from/to a wireless circuit (see FIG. 13 and FIG. 14). For example, the connection point **104a** is a node in a line connecting the other end of the reactance element **102** to the port 1.

(28) The one end of the radiating element **101b** is connected to a port 2 receiving/outputting a signal from/to the wireless circuit. A connection point **104b** (node) between the radiating element **101b** and the susceptance element **103** is positioned between the one end of the radiating element **101b** and the port 2. For example, the connection point **104b** is a node in a line connecting the one end of the radiating element **101b** to the port 2.

(29) FIG. 2 illustrates configuration examples of each of the reactance element **102** and the susceptance element **103**.

(30) As illustrated in FIG. 2A, each of the reactance element **102** and the susceptance element **103** may be configured by an inductor.

(31) As illustrated in FIG. 2B, each of the reactance element **102** and the susceptance element **103** may be configured by a capacitor.

(32) As illustrated in FIG. 2C, each of the reactance element **102** and the susceptance element **103** may be configured by an inductor and a capacitor connected in series.

(33) As illustrated in FIG. 2D, each of the reactance element **102** and the susceptance element **103** may be configured by an inductor and a capacitor connected in parallel.

(34) The reactance element **102** and the susceptance element **103** may have the same configuration or different configurations. For example, both of the reactance element **102** and the susceptance element **103** may be inductors. One of the reactance element **102** or the susceptance element **103** is an inductor, and the other may be a capacitor.

(35) In the decoupling circuit **100** in FIG. 1, the reactance of the reactance element **102** and the susceptance of the susceptance element **103** are appropriately adjusted. This makes it possible to reduce coupling between the first radiating element **101a** and the second radiating element **101b**. In the present embodiment, it is necessary to adjust only two circuit elements in order to reduce coupling. Therefore, adjustment work is easily performable.

(36) Further, when a single inductor or a single capacitor is used as each of the circuit elements as illustrated in FIG. 2A or FIG. 2B, the work to adjust the reactance of the reactance element **102** and the susceptance of the susceptance element **103** is facilitated. In addition, a manufacturing cost can be reduced.

(37) When a configuration in which an inductor and a capacitor are connected in series or in parallel is used as each of the circuit elements as illustrated in FIG. 2C or FIG. 2D, coupling between the first radiating element **101a** and the second radiating element **101b** can be reduced in a plurality of frequency bands.

(38) The inductor or the capacitor illustrated in FIG. 2A to FIG. 2D may include a parasitic inductor, a parasitic capacitor, and an internal resistance.

(39) In a case where the reactance element **102** and the susceptance element **103** are the inductors, a chip component such as a chip inductor or a lead inductor may be used. Further, each of the inductors may be configured by forming a conductor pattern of a substrate in a meandering shape, a spiral shape, or the like. Each of the inductors may be configured using a via of the substrate. Further, a stub may be used as each of the inductors. Each of the inductors may be any of a fixed inductor and a variable inductor.

(40) In a case where the reactance element **102** and the susceptance element **103** are capacitors, a chip component such as a chip capacitor or a lead capacitor may be used. A gap between two conductor patterns formed on the substrate may be used as a capacitor, as with an interdigital capacitor or the like. A metal-insulator-metal (MIM) capacitor configured by a conductor pattern and a base material of the substrate may be used. The capacitor may be any of a fixed capacitor and a variable capacitor. A varactor diode may be used as the variable capacitor.

(41) FIG. 3 is a diagram to explain an example of determining the reactance of the reactance element **102** and the susceptance of the susceptance element **103** to reduce coupling between the first radiating element **101a** and the second radiating element **101b**. In the following, an example of deriving the reactance of the reactance element **102** and the susceptance of the susceptance element **103** by using expressions is described with reference to FIG. 3.

(42) As illustrated in FIG. 3, the reactance of the reactance element **102** is denoted by “X”, and the susceptance of the susceptance element is denoted by “B”. Further, an impedance matrix of the first radiating element **101a** and the second radiating element **101b** as viewed from a to surface in FIG. 3 is represented by the following expression (1). The impedance matrix represents antenna

characteristics of the first radiating element **101a** and the second radiating element **101b**.

[Expression1]

$$(43) \quad [Z] = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} R_{11} + jX_{11} & R_{12} + jX_{12} \\ R_{21} + jX_{21} & R_{22} + jX_{22} \end{bmatrix} \quad (1)$$

(44) In the expression, “Z.sub.11” is an input impedance of the first radiating element **101a**, “Z.sub.22” is an input impedance of the second radiating element **101b**, and “Z.sub.21” and “Z.sub.12” are mutual impedances between the first radiating element **101a** and the second radiating element **101b** and are the same value. Further, “Z.sub.11”, “Z.sub.12”, “Z.sub.21”, and “Z.sub.22” are complex numbers, “R.sub.11”, “R.sub.12”, “R.sub.21”, and “R.sub.22” are real parts, and “X.sub.11”, “X.sub.12”, “X.sub.21”, and “X.sub.22” are imaginary parts. The impedance matrix in the expression (1) is an impedance matrix in a case where it is assumed that nothing is connected to the first radiating element **101a** and the second radiating element **101b**.

(45) An impedance matrix as viewed from a t.sub.1 surface including the first radiating element **101a**, the second radiating element **101b**, and the reactance element **102** is represented by the following expression (2). In the expression, “Z.sub.11+jX” is an input impedance of the first radiating element **101a** including the reactance element **102**, and “X” is the reactance of the reactance element **102**.

[Expression2]

$$(46) \quad [Z^A] = \begin{bmatrix} Z_{11} + jX & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} \quad (2)$$

(47) Further, an admittance matrix is represented by the following expression (3).

[Expression3]

$$(48) \quad [Y^A] = [Z^A]^{-1} = \frac{1}{(Z_{11} + jX)Z_{22} - Z_{12}Z_{21}} \begin{bmatrix} Z_{22} & -Z_{12} \\ -Z_{21} & Z_{11} + jX \end{bmatrix} \quad (3)$$

(49) An admittance matrix as viewed from a t.sub.2 surface including the susceptance element **103** is represented by the following expression (4). The admittance matrix of the expression (4) represents entire characteristics including all of the first radiating element **101a**, the second radiating element **101b**, the reactance element **102**, and the susceptance element **103**. Further, “B” is the susceptance of the susceptance element **103**.

[Expression4]

$$(50) \quad [Y^B] = [Y^A] + \begin{bmatrix} jB & -jB \\ -jB & jB \end{bmatrix} = \begin{bmatrix} Y_{11}^B & Y_{12}^B \\ Y_{21}^B & Y_{22}^B \end{bmatrix} \quad (4)$$

When Y.sub.12.sup.B that is (1, 2) component of [Y.sup.B] is zero, the high-frequency signal input from the port 1 is not output to the port 2. In other words, coupling between the first radiating element **101a** and the second radiating element **101b** is eliminated.

(51) Accordingly, the reactance of the reactance element **102** is adjusted to the reactance “X” represented by the following expression (5), derived from Y.sub.12.sup.B=0, and the susceptance of the susceptance element **103** is adjusted to the susceptance “B” represented by the expression (6). This makes it possible to eliminate or reduce coupling between the first radiating element **101a** and the second radiating element **101b**. In the expression (6), “Inn” represents an imaginary part of a complex number in brackets. Further, Y.sub.12.sup.B=0 is achievable at a plurality of frequencies f. Accordingly, coupling between the first radiating element **101a** and the second radiating element **101b** can be eliminated or reduced in a plurality of frequency bands or in a broad frequency band.

[Expression5]

[Expression6]

$$(52) \quad X = \frac{-(R_{12}^2 + X_{12}^2)R_{21} + (R_{11}R_{12} + X_{11}X_{12})R_{22} + (R_{11}X_{12} - X_{11}R_{12})X_{22}}{R_{12}X_{22} + X_{12}R_{22}} \quad (5) \quad B = \text{Im}\left[\frac{-Z_{12}}{(Z_{11} + jX)Z_{22} - Z_{12}Z_{21}}\right] \quad (6)$$

(53) The reactance “X” of the reactance element **102** and the susceptance “B” of the susceptance element **103** do not necessarily have to be equal to the value derived from the expression (5) and the value derived from the expression (6), respectively. When the reactance “X” of the reactance element **102** and the susceptance “B” of the susceptance element **103** are respectively substantially close to the value derived from the expression (5) and the value derived from the expression (6), coupling between the first radiating element **101a** and the second radiating element **101b** is sufficiently reduced. For example, in fact, when the reactance “X” of the reactance element **102** and the susceptance “B” of the susceptance element **103** are respectively shifted from the value derived from the expression (5) and the value derived from the expression (6) due to influence by solder, wirings, or the like of the circuit, coupling between the radiating elements can be further reduced in some cases. The value close to the value derived from the expression (5) and the value close to the value derived from the expression (6) may be selected from E series circuit constants (standard sequence). The reactance “X” of the reactance element **102** and the susceptance “B” of the susceptance element **103** may be deviated from the value derived from the expression (5) and the value derived from the expression (6), respectively, due to tolerances of the circuit constants, manufacturing errors, or the like.

(54) The susceptance and the reactance are in a reciprocal relationship. Therefore, adjustment of the reactance of the circuit element **102** (reactance element **102**) can be replaced with adjustment of a susceptance of the circuit element **102**. Likewise, adjustment of the susceptance of the circuit element **103** (susceptance element **103**) can be replaced with adjustment of a reactance of the circuit element **103**.

(55) FIG. 4 is a diagram illustrating a decoupling circuit **110** modified from the decoupling circuit **100** in FIG. 1.

(56) In the decoupling circuit **110**, the reactance element **102** is connected between the second radiating element **101b** and the second connection point **104b**. When the reactance “X” of the reactance element **102** and the susceptance “B” of the susceptance element **103** respectively have a value represented by the following expression (7) and a value represented by the following expression (8), coupling between the first radiating element **101a** and the second radiating element **101b** can be made zero.

[Expression7]

[Expression8]

$$(57) \quad X = \frac{-(R_{21}^2 + X_{21}^2)R_{12} + (R_{22}R_{21} + X_{22}X_{21})R_{11} + (R_{22}X_{21} - X_{22}R_{21})X_{11}}{R_{21}X_{11} + X_{21}R_{11}} \quad (7) \quad B = \text{Im}\left[\frac{-Z_{21}}{(Z_{22} + jX)Z_{11} - Z_{21}Z_{12}}\right] \quad (8)$$

(58) As in the decoupling circuit **100** in FIG. 1, the reactance “X” of the reactance element **102** and the susceptance “B” of the susceptance element **103** do not necessarily have to be equal to the value derived from the expression (7) and the value derived from the expression (8), respectively. When the reactance “X” of the reactance element **102** and the susceptance “B” of the susceptance element **103** are respectively substantially close to the value derived from the expression (7) and the value derived from the expression (8), coupling between the first radiating element **101a** and the second radiating element **101b** can be sufficiently reduced.

(59) FIG. 5 illustrates a solid-line graph G1 representing frequency characteristics of coupling between the antennas in the decoupling circuit **100** in FIG. 1, and a dashed-line graph G2 representing frequency characteristics of coupling between antennas in a configuration (comparative example) in which the reactance element **102** and the susceptance element **103** are removed from the decoupling circuit **100** in FIG. 1.

(60) More specifically, the solid-line graph G1 represents coupling between the antennas as viewed from the first connection point **104a** and the second connection point **104b** in a case where, at a center frequency of the operation frequency band, the value derived from the expression (5) is

applied as the reactance “X” to the reactance element **102** and the value derived from the expression (6) is applied as the susceptance “B” to the susceptance element **103**. The dashed-line graph G2 represents coupling between the antennas in a case where the reactance element **102** is not connected to directly connect the first radiating element **101a** and the first connection point **104a**, and the susceptance element **103** is not connected to disconnect the first connection point **104a** and the second connection point **104b**.

(61) As illustrated by the graph G2, in the case where the reactance element **102** and the susceptance element **103** are not connected, coupling between the antennas is large at the center frequency and is -7 dB. In contrast, as illustrated by the graph G1, in the case where the reactance element **102** and the susceptance element **103** are connected, it is possible to reduce coupling between the antennas to -40 dB or less.

(62) FIG. 6 is a diagram to explain frequency characteristics of coupling between the antennas in a case where the reactance “X” of the reactance element **102** is changed from the value derived from the expression (5), in the decoupling circuit **100** in FIG. 1. As the susceptance “B” of the susceptance element **103**, the value derived from the expression (6) is used.

(63) A graph X represents frequency characteristics in a case where the value derived from the expression (5) is used as the reactance “X” of the reactance element **102**. A graph $0.5\times$ represents frequency characteristics in a case where a value that is 0.5 times of the value derived from the expression (5) is used as the reactance “X” of the reactance element **102**. A graph $2\times$ represents frequency characteristics in a case where a value that is two times of the value derived from the expression (5) is used as the reactance “X” of the reactance element **102**. As illustrated in FIG. 6, even in the case where the value that is 0.5 times or two times of the reactance “X” derived from the expression (5) is used, coupling between the antennas at the center frequency is small, namely, -32 dB or less.

(64) FIG. 7 is a diagram to explain frequency characteristics of coupling between the antennas in a case where the susceptance “B” of the susceptance element **103** is changed from the value derived from the expression (6), in the decoupling circuit **100** in FIG. 1. As the reactance “X” of the reactance element **102**, the value derived from the expression (5) is used.

(65) A graph B represents frequency characteristics in a case where the value derived from the expression (6) is used as the susceptance “B” of the susceptance element **103**. A graph $0.5B$ represents frequency characteristics in a case where a value that is 0.5 times of the value derived from the expression (6) is used as the susceptance “B” of the susceptance element **103**. A graph $2B$ represents frequency characteristics in a case where a value that is two times of the value derived from the expression (6) is used as the susceptance “B” of the susceptance element **103**. As illustrated in FIG. 7, even in the case where the value that is 0.5 times or two times of the susceptance “B” derived from the expression (6) is used, coupling between the antennas at the center frequency is small, namely, -32 dB or less.

(66) As can be understood from FIG. 6 and FIG. 7, the reactance “X” of the reactance element **102** and the susceptance “B” of the susceptance element **103** in the decoupling circuit **100** do not necessarily have to be identical to the value derived from the expression (5) and the value derived from the expression (6), respectively. Even when the values deviated from the values derived from the expression (5) and the expression (6) such as values that are 0.5 times or two times of the values derived from the expression (5) and the expression (6) are used, coupling between the antennas can be reduced.

(67) Likewise, the reactance “X” of the reactance element **102** and the susceptance “B” of the susceptance element **103** in the decoupling circuit **110** in FIG. 4 do not necessarily have to be identical to the value derived from the expression (7) and the value derived from the expression (8), respectively. Even when the values separated from the values derived from the expression (7) and the expression (8) such as values that are 0.5 times or two times of the values derived from the expression (7) and the expression (8) are used, coupling between the antennas can be reduced.

(68) FIG. 8 is a diagram to explain an effect of reducing coupling between the first radiating element **101a** and the second radiating element **101b** according to the present embodiment. The values determined by the above-described method are set as the reactance of the reactance element **102** and the susceptance of the susceptance element **103**.

(69) As a result, in a case where a transmission signal is transmitted from the port 1 by using the first radiating element **101a** as illustrated in FIG. 8A, a signal input from the first radiating element **101a** to the second radiating element **101b** through coupling (electromagnetic coupling) and a signal input from the second end of the reactance element **102** (first circuit element) to the one end of the second radiating element **101b** through the susceptance element **103** (second circuit element) are at least partially cancelled. More specifically, when amplitudes of both signals are equal to each other or a difference between the amplitudes of both signals is within an allowable range, and a phase difference is 180 degrees or within an allowable range, both signals are at least partially cancelled. As a result, coupling from the first radiating element **101a** to the second radiating element **101b** is reduced.

(70) Likewise, in a case where a transmission signal is transmitted from the port 2 by using the second radiating element **101b** as illustrated in FIG. 8B, a signal input from the second radiating element **101b** to the first radiating element **101a** through coupling and a signal input from the one end of the second radiating element **101b** to the second end of the reactance element **102** (first circuit element) through the susceptance element **103** (second circuit element) are at least partially cancelled. More specifically, when amplitudes of both signals are equal to each other or a difference between the amplitudes of both signals is within an allowable range, and a phase difference is 180 degrees or within an allowable range, both signals are at least partially cancelled. As a result, coupling from the second radiating element **101b** to the first radiating element **101a** is reduced.

(71) As described above, according to the present embodiment, only two circuit elements are used to reduce coupling between the first radiating element **101a** and the second radiating element **101b**, and the reactance and the susceptance of the two circuit elements are set by the above-described method. This makes it possible to reduce coupling between the first radiating element **101a** and the second radiating element **101b** while simplifying the configuration of the decoupling circuit. Accordingly, it is possible to suppress reduction of the realized gains of the antennas, and to improve the frequency characteristics. Further, in a case where the chip components are used for the circuit elements, the work to adjust the circuit constants (reactance, susceptance, or the like) of the circuit elements is facilitated. Furthermore, it is possible to reduce the manufacturing cost of the decoupling circuit.

Second Embodiment

(72) FIG. 9 is a diagram illustrating a configuration of a decoupling circuit **200** according to a second embodiment. The decoupling circuit **200** further includes a first matching circuit **205a** in addition to the decoupling circuit **100**. The first matching circuit **205a** is provided, which reduces (suppresses) reflection from a first radiating element **201a**. As a result, a loss caused by mismatch can be reduced, which makes it possible to improve communication quality and the like. The first matching circuit **205a** can be configured by, for example, one or more chip components.

(73) FIG. 10 illustrates specific examples of the first matching circuit **205a**. As illustrated in FIG. 10A, the first matching circuit **205a** is, for example, an L-type circuit including a susceptance element **206** and a reactance element **207**. The L-type circuit is connected between a first connection point **204a** and the port 1. One end of the reactance element **207** is connected to the first connection point **204a**, and the other end of the reactance element **207** is connected to the port 1. One end of the susceptance element **206** is connected to the one end of the reactance element **207** on the first connection point **204a** side. The other end of the susceptance element **206** is connected to a reference voltage. As illustrated in FIG. 10B, the one end of the susceptance element **206** in FIG. may be connected to the other end of the reactance element **207**. The reactance element **207** is

configured by a circuit element (fourth circuit element or sixth circuit element), and the susceptance element **206** is configured by a circuit element (third circuit element or fifth circuit element). The circuit element configuring the reactance element **207** and the circuit element configuring the susceptance element **206** may have the same configuration or different configurations.

(74) For example, each of the susceptance element **206** and the reactance element **207** may be configured by the inductor in FIG. 2A described above, or may be configured by the capacitor in FIG. 2B. Each of the susceptance element **206** and the reactance element **207** may be configured by connecting the inductor and the capacitor in series as illustrated in FIG. 2C, or may be configured by connecting the inductor and the capacitor in parallel as illustrated in FIG. 2D. In a case where the reactance element **207** and the susceptance element **206** are the inductors, a chip component such as a chip inductor or a lead inductor may be used, or each of the inductors may be configured, for example, by the formation method using the substrate described above. One chip component including both of the susceptance element **206** and the reactance element **207** may be used.

(75) Further, a plurality of matching circuits **205a** in FIG. 10A or FIG. 10B connected in series may be used as the matching circuit **205a** in FIG. 9. Connecting the plurality of matching circuits **205a** in FIG. 10A or FIG. 10B in series makes it possible to suppress reflection from the first radiating element **201a** in a broad frequency range. The first matching circuit may not be the L-type circuit, and may be configured by a H-type circuit, a T-type circuit, a stub, or the like.

(76) FIG. 11 is a diagram illustrating a modification **210** of the decoupling circuit **200**. The decoupling circuit **210** includes a second matching circuit **205b** connected between a second connection point **204b** and the port 2. The second matching circuit **205b** is provided, which makes it possible to suppress reflection from a second radiating element **201b**. The configuration of the second matching circuit **205b** may be similar to the configuration of the first matching circuit **205a** (see FIG. 10).

(77) FIG. 12 is a diagram illustrating a modification **220** of the decoupling circuit **200**. The decoupling circuit **220** includes both of the first matching circuit **205a** and the second matching circuit **205b**. Both of the first matching circuit **205a** and the second matching circuit **205b** are provided, which makes it possible to suppress reflection from the first radiating element **201a** and reflection from the second radiating element **201b**.

Third Embodiment

(78) FIG. 13 is a diagram illustrating a configuration of a wireless device **300** according to a third embodiment. The wireless device **300** further includes a wireless circuit **308** in addition to the decoupling circuit **220** illustrated in FIG. 12.

(79) For example, the wireless device **300** can radiate and receive electromagnetic waves through a first radiating element **301a** and a second radiating element **301b**, and can perform communication with the other wireless communication device.

(80) In a case where the wireless device **300** performs wireless communication with the other wireless communication device, the wireless circuit **308** generates, as a radio signal, a high-frequency signal used for the wireless communication. The wireless circuit **308** modulates the radio signal, and supplies the modulated signal to the first radiating element **301a** and the second radiating element **301b**, thereby radiating radio waves to a space. Further, radio waves from the other wireless communication device are received by the first radiating element **301a** and the second radiating element **301b**, and high-frequency signals as radio signals are supplied to the wireless circuit **308**. The wireless circuit **308** acquires data from the other wireless communication device by demodulating the supplied high-frequency signals. The wireless device **300** may perform the wireless communication by selecting one of the first radiating element **301a** or the second radiating element **301b** based on a propagation environment and the like, or may perform the wireless communication by multiplying each of the high-frequency signals of the first radiating element **301a** and the second radiating element **301b** by a weight coefficient.

(81) The wireless communication may be performed using one of the first radiating element **301a** or the second radiating element **301b**. FIG. **14** illustrates a configuration example of the wireless device in this case.

(82) FIG. **14** illustrates an example in which the wireless circuit **308** is provided with a switch. A switch **309** is selectively connected to one of the port 1 or the port 2. To use the first radiating element **301a**, the wireless circuit **308** connects the switch **309** to the port 1, whereas to use the second radiating element **301b**, the wireless circuit **308** connects the switch **309** to the port 2.

(83) The wireless device **300** may perform sensing by using the first radiating element **301a** and the second radiating element **301b**. For example, the wireless device **300** irradiates a measurement object with electromagnetic waves radiated from the first radiating element **301a** and the second radiating element **301b**, receives electromagnetic waves reflected by the measurement object by the first radiating element **301a** and the second radiating element **301b**, thereby estimating a position, a shape, a moving speed, and the like of the measurement object.

(84) In a case where the wireless device **300** performs sensing of the measurement object, the wireless circuit **308** generates, as a measurement signal, a high-frequency signal used for the sensing. The wireless circuit **308** radiates radio waves to the space by supplying the measurement signal to the first radiating element **301a** and the second radiating element **301b**. At this time, the wireless circuit **308** may modulate the high-frequency signal or may not modulate the high-frequency signal. The wireless circuit **308** may multiply the high-frequency signal to be supplied to the first radiating element **301a** and the second radiating element **301b** by a weight coefficient. Reflected waves from the measurement object are received by the first radiating element **301a** and the second radiating element **301b**, and the high-frequency signals are supplied to the wireless circuit **308**. The wireless circuit **308** may estimate at least one of the position, the shape, the moving speed, or the like of the measurement object based on at least one of power, a phase, a frequency, or the like of the high-frequency signal received from the first radiating element **301a** and the second radiating element **301b**.

(85) The sensing may be performed using one of the first radiating element **301a** or the second radiating element **301b**. In this case, the wireless circuit **308** may include a switch selectively connected to one of the port 1 or the port 2. A configuration example of the wireless device may be similar to the configuration in FIG. **14**.

(86) Further, the wireless device **300** may generate power by using the electromagnetic waves received by the first radiating element **301a** and the second radiating element **301b**. For example, the wireless circuit **308** rectifies the received electromagnetic waves, stores the rectified electromagnetic waves in a storage battery, or provides the rectified electromagnetic waves to an electronic device as power. The wireless device **300** may use one of the first radiating element **301a** or the second radiating element **301b** to receive the electromagnetic waves, and generate power. In this case, the wireless circuit **308** may include a switch selectively connected to one of the port 1 or the port 2. A configuration example of the wireless device may be similar to the configuration in FIG. **14**.

Applicable Technical Field

(87) The present embodiment is applicable to a wireless device mounted with a plurality of antennas. In particular, in a small wireless device, characteristics of antennas are changed due to proximity of a plurality of antennas, and communication quality and the like of wireless communication are deteriorated by electromagnetic coupling between the antennas. Using the decoupling circuit according to the present embodiment makes it possible to reduce coupling between the antennas, and to improve communication quality and the like. Since the decoupling circuit according to the present embodiment uses only two circuit elements, adjustment of circuit constants is easily performable, and a manufacturing cost can be reduced.

(88) While certain embodiment have been described, these embodiment have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel

embodiments described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions.

(89) The embodiments as described before may be configured as below.

Clauses

(90) Clause 1. Electronic circuitry, comprising: a first radiating element; a second radiating element; a first circuit element including a first end and a second end, the first end being connected to one end of one of the first radiating element or the second radiating element; and a second circuit element connected between the second end of the first circuit element and one end of another of the first radiating element or the second radiating element. Clause 2. The electronic circuitry according to Clause 1, wherein a reactance or a susceptance of the first circuit element and a reactance or a susceptance of the second circuit element are respectively set to values at which a signal input from the one radiating element to the other radiating element through electromagnetic coupling and a signal input from the second end of the first circuit element to the one end of the other radiating element through the second circuit element are at least partially cancelled, and a signal input from the other radiating element to the one radiating element through electromagnetic coupling and a signal input from the one end of the other radiating element to the second end of the first circuit element through the second circuit element are at least partially cancelled. Clause 3. The electronic circuitry according to Clause 1 or 2, wherein each of the first circuit element and the second circuit element includes a coil, a capacitor, series connection of a coil and a capacitor, or parallel connection of a coil and a capacitor. Clause 4. The electronic circuitry according to any one of Clauses 1 to 3, wherein the first circuit element and the second circuit element are chip components. Clause 5. The electronic circuitry according to any one of Clauses 1 to 4, further comprising a first matching circuit connected to the second end of the first circuit element and configured to reduce a reflected wave from the one radiating element. Clause 6. The electronic circuitry according to Clause 5, wherein the first matching circuit at least includes a third circuit element and a fourth circuit element, the fourth circuit element includes one end connected to the second end of the first circuit element, and the third circuit element includes one end connected to the one end or another end of the fourth circuit element, and includes another end connected to a reference voltage. Clause 7. The electronic circuitry according to Clause 5 or 6, wherein the first matching circuit includes one or more chip components. Clause 8. The electronic circuitry according to any one of Clauses 1 to 7, further comprising a second matching circuit connected to the one end of the other radiating element, and configured to reduce a reflected wave from the other radiating element. Clause 9. The electronic circuitry according to Clause 8, wherein the second matching circuit at least includes a fifth circuit element and a sixth circuit element; the sixth circuit element includes one end connected to the one end of the other radiating element, and the fifth circuit element includes one end connected to the one end or another end of the sixth circuit element, and includes another end connected to a reference voltage. Clause 10. The electronic circuitry according to Clause 8 or 9, wherein the second matching circuit includes one or more chip components. Clause 11. The electronic circuitry according to any one of Clauses 1 to further comprising: a first matching circuit connected to the second end of the first circuit element and configured to reduce reflected waves from the one radiating element; and a second matching circuit connected to the one end of the other radiating element and configured to reduce reflected waves from the other radiating element. Clause 12. The electronic circuitry according to Clause 11, wherein the first matching circuit at least includes a third circuit element and a fourth circuit element, the fourth circuit element includes one end connected to the second end of the first circuit element, the third circuit element includes one end connected to the one end or another end of the fourth circuit element, and includes another end connected to a reference voltage, the second

matching circuit at least includes a fifth circuit element and a sixth circuit element, the sixth circuit element includes one end connected to the one end of the other radiating element, and the fifth circuit element includes one end connected to the one end or another end of the sixth circuit element, and includes another end connected to the reference voltage. Clause 13. The electronic circuitry according to Clause 11 or 12, wherein each of the first matching circuit and the second matching circuit includes one or more chip components. Clause 14. The electronic circuitry according to any one of Clauses 1 to 13, further comprising a wireless circuit connected to the second end of the first circuit element and one end of the other radiating element. Clause 15. The electronic circuitry according to Clause 14, further comprising a switch configured to selectively connect the wireless circuit to the second end of the first circuit element or the one end of the other radiating element. Clause 16. The electronic circuitry according to Clause 14 or 15, wherein the wireless circuit is configured to perform sensing of a measurement object by transmitting a measurement signal to the measurement object through at least one of the first radiating element and the second radiating element, and receiving a reflection signal from the measurement object through at least one of the first radiating element and the second radiating element. Clause 17. The electronic circuitry according to any one of Clauses 14 to 16, wherein the wireless circuit performs wireless communication with another wireless communication device by transmitting/receiving a radio signal through at least one of the first radiating element and the second radiating element.

Claims

1. Electronic circuitry, comprising: a first radiating element; a second radiating element; a first circuit element including a first end and second end, the first end being connected to one end of one of the first radiating element or the second radiating element; and a second circuit element connected between the second end of the first circuit element and one end of another of the first radiating element or the second radiating element, wherein no other circuit element is provided between the second circuit element and the one end of the other of the first radiating element or the second radiating element and, a reactance of the first circuit element is in a range of 0.5 to 2 times a value of X in the following equation, $X = \frac{-(R_{12}^2 + X_{12}^2)R_{21} + (R_{11}R_{12} + X_{11}X_{12})R_{22} + (R_{11}X_{12} - X_{11}R_{12})X_{22}}{R_{12}X_{22} + X_{12}R_{22}}$ a susceptance of the second circuit element is in a range of 0.5 to 2 times a value of B in the

following equation, $B = \text{Im}[\frac{-Z_{12}}{(Z_{11} + jX)Z_{22} - Z_{12}Z_{21}}]$ $[Z] = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} R_{11} & +jX_{11} & R_{12} & +jX_{12} \\ R_{21} & +jX_{21} & R_{22} & +jX_{22} \end{bmatrix}$

Z.sub.11 is an input impedance of the first radiating element, Z.sub.22 is an input impedance of the second radiating element, Z.sub.21 and Z.sub.12 each are a mutual impedance between the first radiating element and the second radiating element, R.sub.11, R.sub.12, R.sub.21 and R.sub.22 are real parts, and X.sub.11, X.sub.12, X.sub.21 and X.sub.22 are imaginary parts.

2. The electronic circuitry according to claim 1, wherein a reactance or a susceptance of the first circuit element and a reactance or a susceptance of the second circuit element are respectively set to values at which a signal input from the one radiating element to the other radiating element through electromagnetic coupling and a signal input from the second end of the first circuit element to the one end of the other radiating element through the second circuit element are at least partially cancelled, and a signal input from the other radiating element to the one radiating element through electromagnetic coupling and a signal input from the one end of the other radiating element to the second end of the first circuit element through the second circuit element are at least partially cancelled.

3. The electronic circuitry according to claim 1, wherein each of the first circuit element and the second circuit element includes a coil, a capacitor, series connection of a coil and a capacitor, or parallel connection of a coil and a capacitor.

4. The electronic circuitry according to claim 1, wherein the first circuit element and the second circuit element are chip components.
5. The electronic circuitry according to claim 1, further comprising a first matching circuit connected to the second end of the first circuit element and configured to reduce a reflected wave from the one radiating element.
6. The electronic circuitry according to claim 5, wherein the first matching circuit at least includes a third circuit element and a fourth circuit element, the fourth circuit element includes one end connected to the second end of the first circuit element, and the third circuit element includes one end connected to the one end or another end of the fourth circuit element, and includes another end connected to a reference voltage.
7. The electronic circuitry according to claim 5, wherein the first matching circuit includes one or more chip components.
8. The electronic circuitry according to claim 1, further comprising a second matching circuit connected to the one end of the other radiating element, and configured to reduce a reflected wave from the other radiating element.
9. The electronic circuitry according to claim 8, wherein the second matching circuit at least includes a fifth circuit element and a sixth circuit element; the sixth circuit element includes one end connected to the one end of the other radiating element, and the fifth circuit element includes one end connected to the one end or another end of the sixth circuit element, and includes another end connected to a reference voltage.
10. The electronic circuitry according to claim 8, wherein the second matching circuit includes one or more chip components.
11. The electronic circuitry according to claim 1, further comprising: a first matching circuit connected to the second end of the first circuit element and configured to reduce reflected waves from the one radiating element; and a second matching circuit connected to the one end of the other radiating element and configured to reduce reflected waves from the other radiating element.
12. The electronic circuitry according to claim 11, wherein the first matching circuit at least includes a third circuit element and a fourth circuit element, the fourth circuit element includes one end connected to the second end of the first circuit element, the third circuit element includes one end connected to the one end or another end of the fourth circuit element, and includes another end connected to a reference voltage, the second matching circuit at least includes a fifth circuit element and a sixth circuit element, the sixth circuit element includes one end connected to the one end of the other radiating element, and the fifth circuit element includes one end connected to the one end or another end of the sixth circuit element, and includes another end connected to the reference voltage.
13. The electronic circuitry according to claim 11, wherein each of the first matching circuit and the second matching circuit includes one or more chip components.
14. The electronic circuitry according to claim 1, further comprising a wireless circuit connected to the second end of the first circuit element and one end of the other radiating element.
15. The electronic circuitry according to claim 14, further comprising a switch configured to selectively connect the wireless circuit to the second end of the first circuit element or the one end of the other radiating element.
16. The electronic circuitry according to claim 14, wherein the wireless circuit is configured to perform sensing of a measurement object by transmitting a measurement signal to the measurement object through at least one of the first radiating element and the second radiating element, and receiving a reflection signal from the measurement object through at least one of the first radiating element and the second radiating element.
17. The electronic circuitry according to claim 14, wherein the wireless circuit performs wireless communication with another wireless communication device by transmitting/receiving a radio signal through at least one of the first radiating element and the second radiating element.

